

ASTWalker (Protocol)

language-neutral seam: every signal traverses the AST only through these methods

iter_call_edges()
iter_imports()

iter_control_flow_nodes()
iter_class_methods()

iter_public_api()
iter_function_signatures()

PythonASTWalker

REFERENCE INSTANCE (implemented, this paper)

iter_call_edges ast.Call inside
 FunctionDef bodies

iter_control_flow_nodes ast.If / For / While /
 Try / With / Match

iter_public_api names in __all__,
 resolved via
 ImportFrom

iter_imports ast.Import +
 ImportFrom

iter_class_methods ast.FunctionDef in
 ast.ClassDef body

iter_function_sigs ast.arguments tuple

TreeSitterRustWalker

HYPOTHETICAL (not implemented)

iter_call_edges syn::ExprCall under
 syn::ItemFn

iter_control_flow_nodes syn::ExprIf /
 ExprWhile / ForLoop
 / ExprMatch / ExprTry

iter_public_api pub fn / pub struct
 in lib.rs, +
 pub use re-exports

iter_imports use items + extern
 crate decls

iter_class_methods methods inside impl
 blocks (self recvr)

iter_function_sigs syn::Signature

GoASTWalker

HYPOTHETICAL (not implemented)

iter_call_edges ast.CallExpr inside
 ast.FuncDecl

iter_control_flow_nodes IfStmt / ForStmt /
 SwitchStmt /
 TypeSwitchStmt /
 ReturnStmt /
 RangeStmt

iter_public_api capitalised idents
 in pkg .go files

iter_imports import declarations

iter_class_methods methods with named
 receivers (no class)

iter_function_sigs ast.FuncType

Signal extractors (extract_signals.py)

call walker.iter_*() only — no signal touches ast.* or any language-specific parser API

C06a call_graph_topology

iter_call_edges -> build call graph -> 8-feature reldiff

C06a' call_graph_wl_kernel

iter_call_edges -> Weisfeiler-Lehman kernel (k=4)

C06b import_edge_set

iter_imports / audit_imports -> R1-R5 -> Jaccard

C06c control_flow_histogram

iter_control_flow_nodes -> sum, normalise, cosine

C06d public_api_signature_equivalence

iter_public_api + iter_class_methods -> per-method verdict

C06f per_function_ast_shape

iter_function_signatures + iter_call_edges -> match & dist

Figure 5. The language-neutral AST-walker seam introduced in v2 Phase 1b. Python is the reference instance; Rust and Go instances are stubs documenting the equivalent constructs.